

Module 03: Perception — Feeling the World Before Seeing It

Learning Objectives

1. Explain how the **felt sense** shapes perception from the ground up, demonstrating that all perceptual experience is affectively toned and somatically grounded.
2. Describe **interoceptive contributions to exteroception** — how the body's internal states influence what and how we perceive in the external world.
3. Analyze the role of **precision weighting** in determining whether bodily feelings amplify or attenuate perceptual experience.
4. Recognize that perception in the felt sense framework is never "neutral" observation but always embodied appraisal.

Introduction

You walk into a room and immediately sense something is wrong. Nothing visible has changed — the furniture is in place, the lighting is normal. Yet your body registers a shift: a tightening in the chest, a quickening of breath, a subtle contraction around the eyes. Before your visual system identifies any specific threat, your felt sense has already appraised the situation. Perception, it turns out, does not begin with the eyes. It begins with the body.

Lisa Feldman Barrett's theory of constructed emotion argues that the brain does not first perceive the world neutrally and then add emotional coloring. Rather, interoceptive predictions — the body's ongoing forecast of its own states — fundamentally shape perceptual processing from the earliest stages. What we perceive is always already filtered through the body's felt assessment of its current condition and the situation's implications for its viability. Active inference makes this precise: interoceptive priors constitute a deep layer of the generative model that modulates all subsequent perceptual inference.

Key Concepts

1. Affective Realism: The Body Colors Perception

Barrett and colleagues have demonstrated what they call **affective realism**: the tendency for interoceptive states to be misattributed to properties of the perceived world. When your body is in a state of physiological arousal, neutral faces appear threatening; ambiguous situations seem dangerous. This is not bias in the pejorative sense but the normal operation of embodied perception: the generative model integrates interoceptive predictions with exteroceptive signals, producing a unified perceptual experience in which the world is always already felt.

2. Interoceptive Inference as Perceptual Foundation

Anil Seth has proposed that interoceptive inference constitutes the foundation of all conscious perception. Before the visual cortex constructs a scene, before the auditory cortex parses a sound, the insular cortex is already generating predictions about the body's internal state. These interoceptive predictions provide the **affective context** within which all other perception occurs. In the felt sense framework, this means that perception is fundamentally an embodied assessment: the organism perceives not "what is out there" but "what this situation means for my body."

3. Precision Weighting and Perceptual Salience

Not all perceptual signals are treated equally. The generative model assigns **precision** to different streams of prediction error, amplifying some and attenuating others. When interoceptive precision is high — when the body's signals are weighted as reliable and important — somatic feelings powerfully shape perception. The anxious person perceives threats everywhere because interoceptive alarm signals are given high precision, biasing perceptual inference toward danger. When interoceptive precision is low, as in depersonalization, the world appears flat, distant, unreal — perception stripped of its felt dimension.

4. Somatic Markers in Perceptual Decision-Making

Damasio's somatic marker hypothesis extends directly into perception. When you perceive an object or situation, the body generates a rapid somatic response — a marker — that tags the percept with affective significance. This marker is not added after perception but woven into it: the felt sense of the object includes its emotional valence. In active inference, this integration occurs because the generative model simultaneously predicts both exteroceptive features (shape, color, sound) and interoceptive consequences (arousal, valence, approach/avoidance). Perception is inherently evaluative.

5. The Phenomenology of Embodied Seeing

Merleau-Ponty (1945) insisted that perception is always performed by a body situated in a world. We do not have a "view from nowhere" but always perceive from within our flesh, shaped by our posture, our fatigue, our mood, our history. The felt sense of perception — the qualitative character of seeing this red, hearing that melody — is inseparable from the body's current state. A sunset perceived while deeply fatigued has a different felt quality than the same sunset perceived in a state of vitality. The body does not merely mediate perception; it constitutes it.

Active Inference Connection

In active inference, perception is the process of updating beliefs about hidden causes of sensory signals by minimizing prediction error. The felt sense reveals that these beliefs are never purely about the external world — they always incorporate the body's assessment of its own condition. Interoceptive priors shape the generative model's expectations, meaning that the organism perceives a world that is always already affectively textured. Free energy minimization in perception is thus a joint optimization: the model must simultaneously explain what is happening out there and what is happening in here, integrating exteroceptive and interoceptive streams into a single, somatically grounded perceptual experience.

Experiential Applications

- **Practice — Noticing the Felt Tone of Perception:** Choose a familiar object — a piece of fruit, a photograph, a stone. Hold it and attend not to its visual or tactile features but to the felt quality of perceiving it. Does perceiving this object produce warmth, neutrality, aversion, curiosity? Notice that this felt tone is not separate from the percept but woven into it. You are observing the interoceptive component of your generative model's perceptual inference.
- **Case Study — Phantom Limb Pain and Perceptual Prediction:** After amputation, many patients continue to perceive the missing limb, sometimes in excruciating pain. In active inference terms, the generative model still predicts sensory input from the limb — proprioceptive, tactile, and nociceptive signals that no longer arrive. The discrepancy between predicted and actual signals generates massive prediction errors that the model interprets as pain. Mirror therapy, which provides visual feedback suggesting the limb is present and moving, works by updating the generative model's predictions, reducing the discrepancy and thereby diminishing the phantom pain.

Cross-References

- **Module 02 (Agents):** The agent's felt sense of agency shapes perceptual expectations — we perceive differently when we feel in control versus helpless.
- **Module 04 (Cognition):** Extends embodied perception into the cognitive processes that interpret and elaborate felt perceptual experience.
- **Unit 03 (Intuitive Knowing):** Explores how the perceptual felt sense, accumulated over experience, becomes the foundation of intuitive judgment.

Summary

Perception in the felt sense framework is never neutral observation. It is embodied appraisal — a process in which interoceptive predictions fundamentally shape what and how we perceive. Affective realism, interoceptive inference, precision weighting, and somatic markers all demonstrate that the body's internal states are not afterthoughts to perception but its foundation. Active inference formalizes this insight: the generative model simultaneously

infers external causes and internal consequences, producing a unified perceptual experience in which the world is always already felt. We do not first see and then feel; we feel our way into seeing.